of our discussion. But having developed the essential ideas of special relativity in Chapter 2, we already have the necessary tools for describing more accurately how string theory calms the violent quantum jitters.

In the more precise answer, we rely upon the same core idea as in the rough answer, but we express it directly at the level of strings. We do this by comparing, in some detail, point-particle and string probes. We will see how the extended nature of the string smears out the information that would be obtainable by point-particle probes, and therefore, again, how it happily does away with the ultra-short-distance behavior responsible for the central dilemma of contemporary physics.

We first consider the way in which point particles would interact, if they were actually to exist, and hence how they could be used as physical probes. The most basic interaction is between two point particles moving on a collision course so that their trajectories will intersect, as in Figure 6.5. If these particles were billiard balls they would collide, and each would be deflected onto a new trajectory. Point-particle quantum field theory shows that essentially the same thing happens when elementary particles collide—they scatter off one another and continue on deflected trajectories—but the details are a little different.

For concreteness and simplicity, imagine that one of the two particles is an electron and the other is its antiparticle, the positron. When matter and antimatter collide, they can. annihilate in a flash of pure energy, producing, for example, a photon.9 To distinguish the ensuing trajectory of the photon from the previous trajectories of the electron and positron, we follow a traditional physics convention and draw it with a wiggly line. The photon will typically travel for a bit and then release the energy derived from the initial electron-positron pair by producing another electron-positron pair with trajectories as indicated on the far right of Figure 6.6. In the end, two particles are fired at each other, they interact through the electromagnetic force, and finally they emerge on deflected trajectories, a sequence of events that bears some similarity to our description of colliding billiard balls.

We are concerned with the details of the interaction—specifically, the point where the initial electron and positron annihilate and produce the photon. The central fact, as will become apparent, is that there is an